Supplementary Material for "Reliability-Aware Fusion of Token Evidence for Image-Text Verification in Unified Discrete Diffusion"

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References to sections, tables, equations, and propositions of the main paper are written in words (for example "Section IV-B of the main paper"); labels with an S belong to this supplement. Every number carries a source comment in the LaTeX file, with its `runs/` file and key.

S-I. Preregistration Record

A. Pins

The artefacts of the study are pinned in Table S1. Hashes are SHA-256 of the committed bytes (git LF convention) unless marked. Some working-tree files are CRLF, so a raw hash of a checked-out file can differ from the committed hash without a change of content; both forms are recorded where both were used.

TABLE S1
Hash Set of the Frozen Weighting, Verified by the Scorer at Start

Artefact	SHA-256
Model `runs/P2-LRF-FROZEN-LFIX/model.npz`	`db7fdb4e9f863a18148a4567d396c36a1bba58dc8b041f079492b10466770f37`
Specification `spec.json`	`9d196d7189dc5b6a2d6d80ed694314c6ed323cac59b0300c42c073a6e40db4b5`
Tuning record `hpo.json`	`63cd79442078414726b327eef76e1e1397b3db1b7c3222449705a3f4acdad04f`
Feature code `data.py`	`14421ffb1061947a8408406750e8417282188efeccf07cac3142179fed9f429b`
Model-family code `families.py`	`1c49827d723ea17017f84ba283fcfa681bf24e10c0a0e7d62b6a83064f850a7e`
Fitting code `lrf.py`	`03ed472e1a3cd8bb2dc41f4a59d7b48f1561cd883012ac4434e08a5c37b9d6ec`
Codebook (untracked)	`ff15b9b5b362dc80f8a18f6e852e6e99478c0e8885646735ed16d1d80e3ae50b`
Cached position marginal at slot length 13 (untracked)	`f7744253a1b997b8b9f28d9b91348e8371f1ea1e3564f6c8bfc2f10aa5b42176`
Re-scored design arrays (digest)	`038ded0724b33e4ff132ce9af4eefd7c115bf82fc834777be1e2f4d69cb68e90`
Fresh fold `folds_lrf.json`	CRLF working copy `a545a83ec881988cc81d40c3598fcb4be6292a59d2c835e4cce6e72d3d71f1de`; the committed LF blob has another hash for the same content

B. Audits

The checks named in the main paper as automated audits run in the authors' own pipeline (not independent reviews) are: the plan review (sign-off of the analysis plans); the adversarial review (signatures with conditions on Addenda A2 to A6 and on the build specification); the code, commit-order, and hash audit (pass verdicts at the final audited commits, with delta audits for every later pin; gate-auditor checks of family L, family LG, and the reference encoders with their parity runs; byte-identical regeneration of the committed outputs after every code merge); and the fold and leak checks (sign-off of the fresh fold, check L7). These checks were run by the authors' own pipeline; none is described as independent. No signature says that a method works.

C. Families

Table S2 lists every test family of the paper. Each family has its own level 0.05 and its own Holm correction; there is no error-rate control across families. The paper has seven families, four on COCO and three on Flickr30K; under a global null with independence the chance of at least one false rejection is $1 - 0.95^7 = 0.30$.

D. Dated Amendments

Row A-8 made the self-check of the tokenisation of the fresh-fold stage run with deterministic convolution algorithms after two stops before any item (pin `1871fe7d`). Row A-9 changed one test only (pin `84be25c4`); production code was byte-identical.

TABLE S2
TEST FAMILIES. m_H: HYPOTHESES IN THE FAMILY'S HOLM CORRECTION

Family	n	Test	m_H	Not allowed
L	450	L1 exact one-sided McNemar on noise; then Holm over L3 (JPEG), L4 (splice), two-sided	3	"confirmed" except for L1
L-NI	450	lower 90% bootstrap bound of the clean net change against -0.03	1	a superiority claim on clean images
L-LEN	449	exact one-sided McNemar on noise with length-matched negatives	1	–
LG	300	exact one-sided McNemar, shot, impulse, speckle (severity 3)	3	a claim beyond the three tested noise types
F1	872	exact one-sided McNemar on noise ($\sigma = 0.1$), Flickr30K	1	a claim for another noise level, model, or dataset
F-NI	872	lower 90% bootstrap bound of the clean net change against -0.03, Flickr30K	1	a superiority claim on clean images
F-LEN	872	exact one-sided McNemar on noise with length-matched negatives, Flickr30K	1	–

E. Addendum A7: Transfer Test on Flickr30K

The addendum A7 (commit `76b85e44`) fixed the object (the frozen LRF of Table S1, unchanged), the data, the three families of Table S2, and the wording before any Flickr30K image or caption was downloaded; it was pushed to the code-hosting service before its data were downloaded. Its amendment A7.2(d) (commit `c79d3ed3`), made before any image was scored, added the exclusion of items whose first caption lies outside 5 to 32 tokens, the caption-length rule of the COCO folds: 128 of the 1 000 test images were excluded, leaving 872, above the floor of 800 of A7.6. Nothing else changed. The images and captions of the Flickr30K test split of the Karpathy split (Young et al., 2014; cited in the main paper) were taken from the Hugging Face dataset `nlphuji/flickr30k` at revision `2b239befc81b6e3f035ce6bd52f5f4d60f5625f7`. The dataset card carries no licence field; the dataset's homepage offers the images for non-commercial research and educational purposes under the Flickr Terms of Use and states that its authors do not own the image copyright. Flickr30K images and captions are therefore not redistributed: the repository stores image identifiers and hashes, never an image or a caption, and the per-item arrays are untracked with a committed SHA-256 manifest. The SHA-256 of the scored predictions is `82f0a24832110a5ee301f6b19587eac2df09f733d470101a9579037845111fd8`.

F. Time Input of the Released Configuration

The main paper (Section III) states that UniDisc's denoiser has no time input in the released configuration. Table S3 gives the evidence, read in the authors' code at commit `3ee3e47a9e64bac2fdca59957726f14723371ced` of the UniDisc repository and in the key names of the checkpoint snapshot `2b344ca856bd108b52a2d022a3463dace131249f` (a code reading, not a reproduction). The statement concerns this configuration and checkpoint, not UniDisc in general. Because the loader is non-strict, a key check was run: all 342 checkpoint tensors match the 342 parameters of the denoiser backbone, with no missing key, no unexpected key, and no shape mismatch. The check does not cover the image tokeniser, which is loaded by another path, and key equality does not show that the values are the intended final weights; how often whole-modality masks occurred in training cannot be recovered from the code.

TABLE S3
EVIDENCE FOR THE ABSENCE OF A TIME INPUT (PATHS RELATIVE TO THE UNIDISC REPOSITORY UNLESS MARKED)

Claim	Evidence (file:line)
Time conditioning is off; absorbing-state diffusion, parameterisation `subs`; no experiment overrides it	`configs/config.yaml:12,14,15`
The project composes the official demonstration experiments and loads the interleaved checkpoint	project files `runs/env-2026-10-14/scripts/unidisc_setup.py:17,30-35`, `exp001/unidisc_net.py:46-47`
The official inference experiment sets a null noise level; the samplers pass `None`	`configs/experiments/large_scale_high_res_interleaved_inference.yaml:30-31`; `model_eval.py:2057,2086,3019,3060,3186`
The forward pass accepts a null noise level only when this is allowed	`model.py:660-662,691,729`
No timestep embedder and no time modulation without time conditioning	`models/dit.py:1105,1186-1188,1377-1379`, `models/dit.py:1064-1090`
A stored scaling flag is never read	`configs/model/extra_large.yaml:8`; `models/dit.py:1290`
No time-embedding or time-modulation keys; "sigma" absent from the checkpoint	key names read from the checkpoint pickle
Raw logits are returned before the step that copies observed positions	`model.py:784-788`

G. Full List of Deviations and Disclosures

1) The L1 stage stopped twice at its tokenisation self-check before any item (directories `P2-LRF-L1.try1`, `.try2`); row A-8 moved the pin to `1871fe7d`. Residual: the self-check certifies the layout under the deterministic backend, while the items ran with autotuning on as in the design run, and a fresh tokenisation can differ by up to about 1.5% of tokens between processes.
2) The scorer's pin file was written at 22:42 while L1 was running; every value in it was copied from records committed before L1 (the hash set of the object commit).
3) Row A-9 (pin `84be25c4`) changed one test only.
4) The environment lock files of the design-run, re-score, L1, and LG stages are empty; the environment was recorded after the runs (`runs/P2-ENV/pip-freeze-2026-10-30.txt`); the commit of the corruption package was verified from the environment.
5) Item arrays are untracked; SHA-256 manifests are committed.
6) Hashes appear in CRLF and LF forms; both refer to the same content.
7) The parity files were moved unchanged from a wrong output directory; no hash was taken before the move, and audit-time SHA-256s are recorded in the reference-encoder audit.
8) The field `pixel_parity: not run` of the main reference-encoder file is superseded by the two parity runs (S-IV).
9) The download of the reference encoders was 5.2 GB against 2 to 4 GB quoted to the first author.
10) Decision D0094: what had been seen before the re-freeze (main paper, Section IV-A).
11) "No output was read before its run was complete" rests on the operations log.
12) The exploratory ablation (S-V) was added at writing; no ablation was preregistered.
13) The denoiser comparison (S-V, plan E1 in `research/paper2/tnnls/EXPLORATORY-PLAN-BASELINES-2026-10-30.md`, commit `83ac79f8`) was planned in a file committed before the run, and the run was made once, on the fresh-fold images, after the confirmatory results were known. The environment lock file of the denoiser run is empty, as for the earlier stages; the environment record of item 4 is dated before that run.
14) The Dempster-combination row (research/paper2/tnnls/EXPLORATORY-PLAN-EVIDENTIAL-2026-10-31.md) and the Gaussian-noise severity sweep (research/paper2/tnnls/EXPLORATORY-PLAN-SEVERITY-2026-10-31.md, Table S14) were planned in files committed before their runs, after the confirmatory results were known; the environment lock file of the sweep run is empty, as for the denoiser run.
15) 2026-10-31: this paper reports one of two preregistered studies; the other is not reported here; the methods state its negative primary result without its numbers. The analysis addendum (A6.12) had placed a sentence on the other study in this paper's abstract, so this is a deviation from it.
16) 2026-10-31: the title was shortened from the title preregistered for the confirmed branch (A6.12, branch A); the claim is unchanged.
17) 2026-10-31: the abstract contains no numbers; they are in the Results. This departs from the sentence templates of A6.12, which carried the numbers in the abstract; no claim was added.
18) 2026-10-31: amendment (d) of Addendum A7 (commit `c79d3ed3`), made before any image was scored: items whose first caption lies outside 5 to 32 tokens were excluded, the caption-length rule of the COCO folds, because the pinned runner stops a fold in which more than 10% of items break it; 128 of the 1 000 test images were excluded, leaving 872 (floor 800). The as-written build is kept as a record.
19) 2026-10-31: addendum A6 had stated that no further confirmatory study would be added; addendum A7 (commit `76b85e44`), written after the COCO results were known, added one confirmatory test of the same frozen object on Flickr30K, in answer to a supervisor's comment and an internal review. Addendum A6 and its results were not changed.

S-II. Exploratory and Descriptive Material

A. Every Quantity Reported With Each Preregistered Test

Table S4 gives, for every preregistered test, every quantity that the analysis plan reports with each test. These are preregistered outputs, not exploratory material.

B. Item-Level Margins on the Fresh Fold (Descriptive)

Figure S1 shows the discordant items of L1 and of the clean cell by margin. It is descriptive, and superiority on clean images was not a preregistered hypothesis.

C. Exploratory and Descriptive Material

Everything in the rest of this section is exploratory or descriptive. The design-item numbers below are post hoc: they set the preregistered sample-size rule only and are not a performance estimate of the frozen object, which has no out-of-sample estimate on the design items.

TABLE S4
EVERY PREREGISTERED TEST WITH ALL REPORTED QUANTITIES

Test	n	b	c	n_d	Acc. none	Acc. LRF	Net	Decision
L1: noise, $\sigma = 0.1$	450	58	27	85	0.733	0.802	**0.069**	**confirmed (one-sided)**
90% [0.036, 0.102]; 95% [0.029, 0.109]; Clopper–Pearson 95% of b/n_d [0.572, 0.779]; exact p: one-sided 0.0005, two-sided 0.001; mid-p: one-sided 0.0004, two-sided 0.0007; Holm p not applicable								
L3: JPEG, quality 10	450	27	20	47	0.849	0.864	0.016	not rejected
90% [−0.009, 0.040]; 95% [−0.013, 0.044]; Clopper–Pearson 95% of b/n_d [0.422, 0.717]; exact p: one-sided 0.191, two-sided 0.382; mid-p: one-sided 0.156, two-sided 0.312; Holm p 0.382								
L4: splice, 128 px	450	24	10	34	0.896	0.927	**0.031**	**rejected (Holm)**
90% [0.009, 0.053]; 95% [0.007, 0.058]; Clopper–Pearson 95% of b/n_d [0.525, 0.849]; exact p: one-sided 0.012, two-sided 0.024; mid-p: one-sided 0.008, two-sided 0.017; Holm p 0.049								
L-NI: clean images	450	27	13	40	0.887	0.918	0.031	non-inferior: lower 90% bound 0.009 above −0.03
90% [0.009, 0.053]; 95% [0.004, 0.058]; Clopper–Pearson 95% of b/n_d [0.509, 0.814]; exact p: one-sided 0.019‡, two-sided 0.038‡; mid-p: one-sided 0.014‡, two-sided 0.028‡; Holm p not applicable								
L-LEN: noise, length-matched	449	67	33	100	0.708	0.784	**0.076**	**rejected (one-sided)**
90% [0.040, 0.111]; 95% [0.033, 0.120]; Clopper–Pearson 95% of b/n_d [0.569, 0.761]; exact p: one-sided 0.0004, two-sided 0.0009; mid-p: one-sided 0.0003, two-sided 0.0006; Holm p not applicable								
LG: shot noise	300	43	15	58	0.673	0.767	**0.093**	**rejected (Holm, one-sided)**
90% [0.053, 0.137]; 95% [0.047, 0.143]; Clopper–Pearson 95% of b/n_d [0.610, 0.847]; exact p: one-sided 0.0002, two-sided 0.0003; mid-p: one-sided 0.0001, two-sided 0.0002; Holm p 0.0005								
LG: impulse noise	300	35	20	55	0.700	0.750	**0.050**	**rejected (Holm, one-sided)**
90% [0.010, 0.090]; 95% [0.000, 0.100]; Clopper–Pearson 95% of b/n_d [0.496, 0.762]; exact p: one-sided 0.029, two-sided 0.058; mid-p: one-sided 0.022, two-sided 0.044; Holm p 0.029								
LG: speckle noise	300	32	14	46	0.697	0.757	**0.060**	**rejected (Holm, one-sided)**
90% [0.023, 0.097]; 95% [0.017, 0.103]; Clopper–Pearson 95% of b/n_d [0.542, 0.823]; exact p: one-sided 0.006, two-sided 0.011; mid-p: one-sided 0.004, two-sided 0.008; Holm p 0.011								

b: items LRF answers correctly and the unweighted rule incorrectly; c: the reverse; $n_d = b + c$; net $= (b - c)/n$. Intervals: paired percentile bootstrap over items (10 000 resamples). ‡ L-NI: p-values against zero net change for completeness only; the preregistered test of L-NI is the lower 90% bound against the margin −0.03, and superiority on clean images was not preregistered. Impulse noise: the two-sided p and the lower 95% bound 0.0 are stated in the main paper; no two-sided claim is made. Bold: null rejected at the preregistered level.

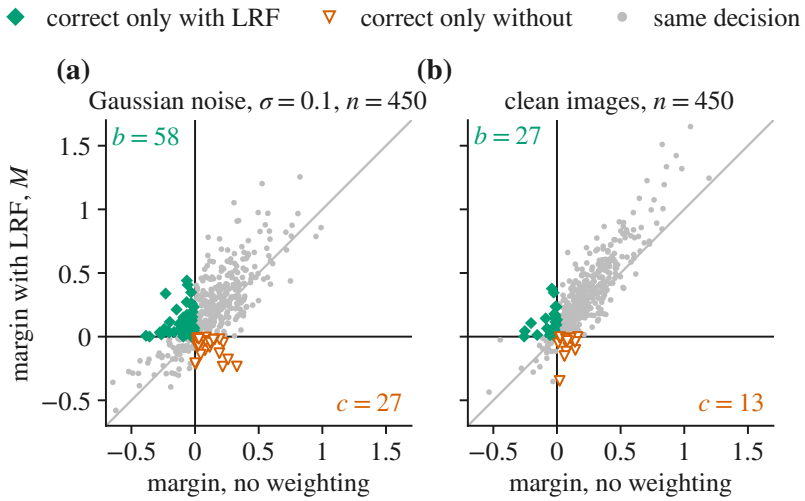


Fig. S1. Item-level margins on the fresh fold (descriptive; no test). Each point is one of the $n = 450$ items: horizontal, the margin without weighting, S/m; vertical, the LRF margin M (5) of the main paper. A decision is correct when its margin is positive (ties wrong), so the upper-left quadrant holds the b items correct only with LRF (filled diamonds) and the lower-right the c items correct only without weighting (open triangles); grey: same decision; line: equal margins. (a) Gaussian noise, standard deviation 0.1 (L1). (b) Clean images (L-NI), same items; descriptive only, and superiority on clean images was not a preregistered hypothesis. Counts b, c as in Table I of the main paper.

Sample-size rule: The rule (Addendum A6, A6.3) sets Δ_{plan} to half of a cross-fitted design-item difference and takes the smallest n on a step-10 grid with exact one-sided McNemar power at least 0.8 at level 0.05, bounded to $[400, 533]$; its inputs and output are in Table S5.

TABLE S5
INPUTS AND OUTPUT OF THE SAMPLE-SIZE RULE (POST HOC DESIGN-ITEM CROSS-FIT; SETS THE RULE ONLY)

Quantity	Value
Cross-fitted design items, noise: b, c, n (5 outer, 3 inner folds)	86, 28, 500
Δ_{cf}; discordance π_{cf}; $\Delta_{\text{plan}} = \Delta_{\text{cf}}/2$	0.116; 0.228; 0.058
n_{req}; n_L	450; 450
Power at n_L for Δ_{plan}, one-sided 0.05; 0.025	0.803; 0.704

Fresh-fold construction: The fold builder read every image set of the project and the seven SugarCrepe files (1 560 images) and excluded their union; images without instance annotations were not eligible. The primary wrong caption follows the derangement rule of the project's fold code; the length-matched wrong caption has the true caption's token count in 531 of 533 items and differs by one token in the other two.

Length strata of L1 (descriptive): By length stratum the L1 net change was 0.038 when the true caption was shorter than the wrong one (184 items), 0.140 when equal (43), and 0.081 when longer (223); longer minus shorter was 0.043 (95% interval −0.038 to 0.123).

Drop of the unweighted rule under the other noise types (descriptive): On the first 300 fold items the unweighted rule lost 0.217 (95% interval 0.163 to 0.270) under shot noise, 0.190 (0.143 to 0.240) under impulse noise, and 0.193 (0.143 to 0.247) under speckle noise, relative to clean images.

Cost: Table S6 gives the cost of the fresh-fold stages; the unweighted rule is the only reference.

TABLE S6
COST PER ITEM OF THE FRESH-FOLD STAGES

Method (stage)	Items	Network rows per item	Compute time per item, s	Peak GPU memory, GB
NONE, all four conditions (fresh-fold pass; the same pass stores the inputs of LRF)	450	67	8.0 (7.9–8.4)	8.2
NONE, three other noise types (LG pass)	300	42	5.0 (5.0–5.1)	8.2
LRF: weights and decision from stored arrays (CPU)	–	not measured	not measured	not measured

One laptop GPU (12 GB). Network rows: median (range); time: median (quartiles) of the busy time per item, excluding duty-cycle pauses; memory: maximum over items. Counted per scored image, the unweighted sum needs 2 passes and LRF $2 + 9 = 11$ (eight complementary-mask passes and one observed-position pass; the cached null for D, the texture and the codebook-neighbour features need none); its weights and decision are then computed from the stored arrays on the CPU: 6.5 s for all 1 800 image-and-condition decisions of the fresh fold (3.6 ms each, start-up included; runs/P2-LRF-TIMING/scorer_timing.json). Source: runs/P2-LRF-RESCORE/items.jsonl (n_rows, median 36 per item for four conditions), runs/P2-LRF-L1/config.json (K), runs/P2-LRF-L1/items.jsonl, runs/P2-LRF-LG/items.jsonl.

D. Weight-Model Search, Caption-Slot Checks, and the Full Feature Table (Main Paper, Section IV-B)

Nine configurations of $(Q, \lambda,$ learning rate) were drawn at random and pruned by synchronous successive halving (factor 3; 4, 13, and 40 epochs) on three inner folds of the 500 design items. On five items under four conditions the caption-masked scores were identical whether the slot was built from the true caption, the wrong caption, or padding, and the scorer reproduced the stored values at the true length (maximum absolute difference 0 in both checks). Any non-finite feature, weight, or margin stops the scorer and is treated as a pipeline defect, never as a wrong decision. Table S7 gives every feature by name and formula; Fig. 2 of the main paper groups them.

Fig. 5 of the main paper shows the learned shape of the weight for the six of the 35 features with the largest learned effect range.

TABLE S7
INPUT FEATURES OF THE FROZEN WEIGHT MODEL, BY NAME, GROUP, AND FORMULA

Group	Count	Features	Definition
token scores	7	`B`, `D`, `q`, `s3`, `tbv`, `ltex`, `nbr`	B_j: log-probability of the observed code at token j with the tokens of j's group masked, the other image tokens visible and the caption masked; D_j: the same with every token masked; $r_j = B_j - D_j$; `s3` $= 1 - \max_y p(y)$ of the pass that gives B_j; `tbv`: log-probability of the observed code from one raw-logit pass with all image tokens visible and the caption masked; `ltex` $= \log(1 + v_j)$, v_j the pixel variance of the 16×16 patch of token j; `nbr`: mean L_2 distance of the codebook vector of token j to those of its eight neighbours
position	3	`row`, `col`, `rad`	`row`, `col`: grid row and column of the token divided by 31; `rad` $= \sqrt{(\text{row} - \frac{1}{2})^2 + (\text{col} - \frac{1}{2})^2}$
neighbourhood	8	`nb3mean_q`, `nb3mean_B`, `nb3mean_s3`, `nb3mean_tbv`, `nb3mean_ltex`, `nb3mean_nbr`, `nb3min_q`, `nb5mean_q`	`nb3mean_`k: mean of score k over the eight neighbours (edge-aware, token excluded); `nb3min_q`: their minimum of r; `nb5mean_q`: mean of r over the 5×5 neighbourhood without the token
within-image rank	4	`rank_q`, `rank_B`, `rank_s3`, `rank_tbv`	percentile rank of the score among the 1 024 tokens of the image, in $[0, 1]$
image mean	6	`itemmean_q`, `itemmean_B`, `itemmean_s3`, `itemmean_tbv`, `itemmean_ltex`, `itemmean_nbr`	mean of the score over the 1 024 tokens of the image
image standard deviation	6	`itemsd_q`, `itemsd_B`, `itemsd_s3`, `itemsd_tbv`, `itemsd_ltex`, `itemsd_nbr`	standard deviation of the score over the 1 024 tokens of the image
image z-score	1	`itemz_q`	$(r_j - \bar{r})/(\text{sd}(r) + 10^{-6})$ within the image
Total	35		

There are 35 features in seven groups; names, order, and counts as in runs/P2-LRF-FROZEN-LFIX/spec.json (features). B_j and D_j use the notation of Section III of the main paper. Source of the formulas: analysis_posthoc/learned_ceiling/data.py, functions feature_names and engineer.

S-III. SYMBOL CROSSWALK

S-IV. REFERENCE ENCODERS AND PIXEL PARITY

The reference encoders are openai/clip-vit-large-patch14 at revision `32bd64288804d66eefd0ccbe215aa642df71cc41` and google/siglip-so400m-patch14-384 at revision `9fdffc58afc957d1a03a25b10dba0329ab15c2a3`, scored with the CLIP cosine score and the SigLIP pairwise logit; the decision is the score of the true caption above that of the wrong caption, with ties counted as wrong. They are run on the same first 450 fold items (300 for the three severity-3 cells) on images regenerated with the runner's corruption functions and seeds; no caption exceeds either text encoder's length limit. Pixel parity: the UniDisc tokeniser's re-encoding of the regenerated images agreed with the stored codes on at

TABLE S8
CROSSWALK FROM PREREGISTRATION AND CODE NAMES TO THE SYMBOLS OF THE PAPER

Preregistration or code	Paper
`acc_a`, `acc_b` in `a6_analysis.json`	accuracy with LRF, accuracy without weighting
`knots`, `ridge`, `lr`, `T` in `spec.json`	Q, λ, learning rate, T
$L_{\mathrm{FIX}} = 13$	caption-slot length 13
`y_lrf_L`, `y_none_L`	L-LEN decisions with the length-matched wrong caption
feature `q` and its derived names in `spec.json`	$r_j = B_j - D_j$
pixel variance in `data.py`	v_j
`sigmoid` in `spec.json`	sigm, the logistic function of the weight model

least 0.989 of tokens for the clean, JPEG, noise, and splice conditions and at least 0.986 for the clean and severity-3 conditions (five items each; threshold 0.98). These two runs supersede the field `pixel_parity: not run` of the main reference-encoder file, which records only that the main run was made without the parity option; the parity files were moved unchanged from a wrong output directory, and audit-time hashes are recorded. Agreement below one is expected from process-to-process tokenisation differences (S-I, deviation 1); the control detects gross pixel errors, not single-pixel ones. No committed run recorded a timing for the encoders, so their cost per item is not reported.

S-V. EXPLORATORY ABLATION AND COMPARISON

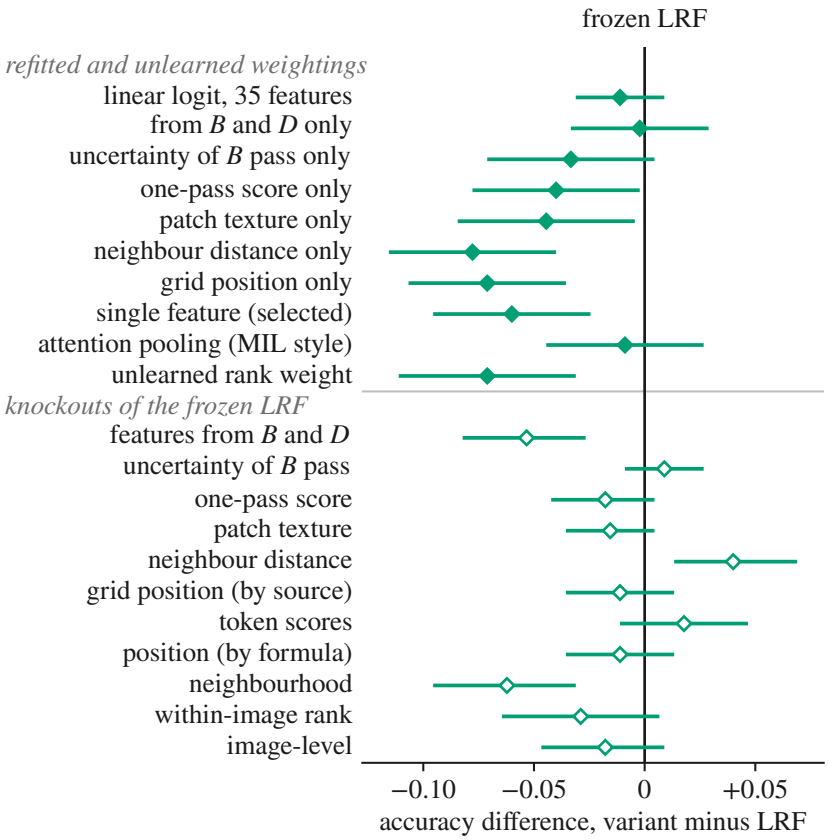


Fig. S2. Exploratory ablation (formerly in the main text; added after the confirmatory results; intervals only, no test): paired accuracy difference, variant minus the frozen LRF, on the L1 noise cell ($n = 450$), with the 95% paired percentile-bootstrap interval (10 000 resamples). A positive value favours the variant. Upper block: weightings refitted with LRF's protocol on the 500 design items only, and one unlearned rank weight. Lower block: knockouts, the frozen LRF with one feature group set to its design-item mean. Intervals are marginal, dependent, and unadjusted (the tables of this section give the other cells).

Label. Exploratory; added after the confirmatory results. No ablation was preregistered. The plan (research/paper2/tnnls/ABLATION-PLAN-2026-10-30.md) was written and then checked by an automated statistical review, and the variant list was committed before any variant was fitted or scored. *Disclosure.* When the plan was written, its author had seen the fresh-fold results of the frozen LRF and of the unweighted rule, an exploratory cross-fit of four learned-weighting

families on the design items, and the frozen LRF's shape functions (Fig. 5 of the main paper); the variant list was fixed after the fresh LRF and unweighted results had been seen, and no variant was fitted or scored before the list was committed. Every listed variant is reported; no variant was selected and scored again, and the tested object stays the frozen LRF.

Variants: All fitted variants use LRF's protocol with the stated change: the same objective, optimiser, nine-configuration search with successive halving, three inner folds over the 500 design items, selection by the inner-fold objective, and one final seed. A-LIN is a linear logit on the 35 features (no hinge terms). A-S-BD, A-S-s3, A-S-tbv, A-S-ltex, A-S-nbr, and A-S-pos are the same additive model on the features computed from one source only: B and D (14 features), s_3 (5), tbv (5), ltex (4), nbr (4), and grid position (3). A-SF is a single-feature weight (feature `rad`, chosen from 35 candidate features by the inner-fold objective on the design items, a selection budget 35 times that of the frozen LRF; the selection used no fresh item). A-MIL is attention-weighted pooling in the style of gated-attention multiple-instance learning (Ilse et al., 2018; cited in the main paper) over the same 35 features, fitted with the same objective, design items, and tuning budget; the evidence Δ_j takes the place of the instance classifier. A-UNL is an unlearned rank weight on s_3, $w_j = 1 - \mathrm{rank}(s_{3,j})$; the direction and the rank transform were fixed by the plan's author, who had seen the frozen LRF's shape functions; they were not read from an accuracy. The knockouts evaluate the frozen LRF with one group set to its design-item mean: by source (KO-S-BD, KO-S-s3, KO-S-tbv, KO-S-ltex, KO-S-nbr, KO-S-pos) and by formula type (KO-F-token, KO-F-position, KO-F-neighbourhood, KO-F-rank, KO-F-item). Setting a group to its mean gives inputs that the model did not see together in fitting; an interval away from zero shows that the frozen model's decisions depend on those inputs, not that the group alone detects anything. KO-S-pos and KO-F-position act on the same three features, so their rows are identical.

Intervals: Each interval is a marginal 95% interval. In all, 336 intervals were computed from the same images (variants × cells × two contrasts); they are positively dependent and no familywise or false-discovery adjustment was made. If every true difference were zero, about 5% of them (about 17) would still exclude zero, and the count of intervals that exclude zero is not a calibrated tally because of the dependence. An interval that excludes zero is therefore a descriptive observation, not a finding; patterns across rows and cells are described, single intervals are not claims. The intervals cover item sampling only: refit-seed variation and design-item sampling are not in them. Intervals are paired percentile bootstrap intervals with 10 000 item resamples, one shared index matrix per cell; differences are variant minus reference, so a positive value favours the variant, and an interval that contains zero does not show that the difference is zero. The 300 images of the LG cells are the first 300 images of the L1 cells, so the eight cells are not independent samples; the cells have 450 (L1) and 300 (LG) items.

Controls: Recomputed from the stored arrays, the unweighted rule and the frozen LRF reproduced the decisions of the preregistered analysis on all 3 000 item-cells (0 mismatches each), and the knockout of all 35 features (KO-ALL, an exact-equality control with no interval) reproduced the unweighted rule on every item-cell. An LRF refit by the ablation harness at seed 0 decided every fresh item as the frozen LRF did.

Results: Tables S9 and S10 give every variant against the frozen LRF and Table S11 against the unweighted rule. Under Gaussian noise (L1) the intervals against LRF contained zero for A-LIN, A-S-BD, A-S-s3, and A-MIL, and lay below zero for A-S-tbv, A-S-ltex, A-S-nbr, A-S-pos, A-SF, and A-UNL; among knockouts, KO-S-BD and KO-F-neighbourhood lay below zero and KO-S-nbr above zero, and every other knockout interval contained zero. In other cells, intervals against LRF that lay above zero were those of A-LIN under JPEG (0.020 [0.004, 0.038]), A-S-BD under impulse noise (0.043 [0.003, 0.083]), and KO-S-s3 under JPEG (0.011 [0.002, 0.022]); this comparison was added after the confirmatory results and is not a test. The tables give every other cell, row by row, without a summary. The unlearned rank weight on s_3 had a paired difference from the unweighted rule of −0.002 [−0.024, 0.020] and from LRF of −0.071 [−0.111, −0.031] under Gaussian noise. Across the three refit seeds, the range of fresh-fold accuracy was at most 0.030 in every variant and cell (Table S12). Seed 0 is the pre-fixed seed that LRF used, not the best of three; the extra seeds repeat only the final fit, not the configuration search or the choice of design items, so they understate refit variation. The seed-spread rows are refits, not the frozen LRF.

Denoising before scoring (exploratory, plan E1): The denoise rows of Table S13 come from runs/P2-LRF-DN: the 450 fresh-fold images, the same captions and the same stored corruption seeds as the L1 run, Gaussian noise with $\sigma = 0.1$ and JPEG quality 10. Each corrupted image was denoised with OpenCV `fastNlMeansDenoisingColored` ($h = h_{\mathrm{color}} = 25.5$, template window 7, search window 21; the setting was fixed in advance and no other was tried) and scored by the unweighted sum (denoise, then unweighted sum) and by the frozen LRF (denoise, then LRF). LRF was not trained on denoised images, so the second arm applies the frozen weights outside their training distribution. Under noise the accuracies were 0.733 (unweighted sum, undenoised), 0.802 (LRF, undenoised), 0.844 (denoise, then unweighted sum) and 0.856 (denoise, then LRF). The denoised unweighted sum exceeded the undenoised unweighted sum by 0.111 (95% interval 0.067 to 0.156; 80 items right only after denoising, 30 right only before) and LRF by 0.042 (0.002 to 0.082; 53 against 34); the denoised LRF exceeded LRF by 0.053 (0.011 to 0.096; 61 against 37), and the two denoised arms differed by 0.011 (−0.020 to 0.044; 30 against 25). Under JPEG the accuracies were 0.849, 0.864, 0.853 and 0.813 in the same order. Against LRF the difference was −0.011 (−0.044 to 0.024; 29 against 34) for the

TABLE S9
Exploratory Ablation (Added After the Confirmatory Results): Accuracy and Difference From the Frozen LRF, L1 Cells

	clean ($n = 450$)		JPEG 10 ($n = 450$)		noise 0.1 ($n = 450$)		splice 128 ($n = 450$)	
Row	Acc.	d vs LRF [95 %]	Acc.	d vs LRF [95 %]	Acc.	d vs LRF [95 %]	Acc.	d vs LRF [95 %]
NONE (reference)	0.887	–	0.849	–	0.733	–	0.896	–
LRF (reference)	0.918	–	0.864	–	0.802	–	0.927	–
A-LIN	0.920	0.002 [−0.013, 0.018]	0.884	0.020 [0.004, 0.038]	0.791	−0.011 [−0.031, 0.009]	0.931	0.004 [−0.009, 0.018]
A-S-BD	0.922	0.004 [−0.016, 0.024]	0.871	0.007 [−0.018, 0.031]	0.800	−0.002 [−0.033, 0.029]	0.911	−0.016 [−0.038, 0.004]
A-S-s3	0.893	−0.024 [−0.051, 0.002]	0.853	−0.011 [−0.040, 0.018]	0.769	−0.033 [−0.071, 0.004]	0.889	−0.038 [−0.062, −0.016]
A-S-tbv	0.913	−0.004 [−0.029, 0.020]	0.856	−0.009 [−0.038, 0.020]	0.762	−0.040 [−0.078, −0.002]	0.904	−0.022 [−0.044, 0.000]
A-S-ltex	0.896	−0.022 [−0.049, 0.004]	0.840	−0.024 [−0.056, 0.004]	0.758	−0.044 [−0.084, −0.004]	0.893	−0.033 [−0.058, −0.011]
A-S-nbr	0.900	−0.018 [−0.042, 0.007]	0.873	0.009 [−0.016, 0.033]	0.724	−0.078 [−0.116, −0.040]	0.898	−0.029 [−0.053, −0.007]
A-S-pos	0.902	−0.016 [−0.044, 0.013]	0.864	0.000 [−0.029, 0.029]	0.731	−0.071 [−0.107, −0.036]	0.891	−0.036 [−0.064, −0.007]
A-SF	0.893	−0.024 [−0.053, 0.002]	0.840	−0.024 [−0.056, 0.007]	0.742	−0.060 [−0.096, −0.024]	0.891	−0.036 [−0.064, −0.007]
A-MIL	0.864	−0.053 [−0.082, −0.024]	0.838	−0.027 [−0.056, 0.002]	0.793	−0.009 [−0.044, 0.027]	0.851	−0.076 [−0.102, −0.049]
A-UNL	0.896	−0.022 [−0.044, 0.000]	0.838	−0.027 [−0.058, 0.004]	0.731	−0.071 [−0.111, −0.031]	0.891	−0.036 [−0.060, −0.013]
KO-S-BD	0.909	−0.009 [−0.027, 0.009]	0.836	−0.029 [−0.053, −0.004]	0.749	−0.053 [−0.082, −0.027]	0.907	−0.020 [−0.038, −0.002]
KO-S-s3	0.913	−0.004 [−0.020, 0.011]	0.876	0.011 [0.002, 0.022]	0.811	0.009 [−0.009, 0.027]	0.918	−0.009 [−0.020, 0.000]
KO-S-tbv	0.904	−0.013 [−0.031, 0.004]	0.880	0.016 [−0.002, 0.036]	0.784	−0.018 [−0.042, 0.004]	0.900	−0.027 [−0.044, −0.009]
KO-S-ltex	0.911	−0.007 [−0.022, 0.007]	0.822	−0.042 [−0.062, −0.024]	0.787	−0.016 [−0.036, 0.004]	0.916	−0.011 [−0.027, 0.004]
KO-S-nbr	0.927	0.009 [−0.011, 0.029]	0.862	−0.002 [−0.024, 0.020]	0.842	0.040 [0.013, 0.069]	0.924	−0.002 [−0.022, 0.018]
KO-S-pos	0.913	−0.004 [−0.020, 0.011]	0.869	0.004 [−0.011, 0.020]	0.791	−0.011 [−0.036, 0.013]	0.918	−0.009 [−0.022, 0.002]
KO-F-token	0.909	−0.009 [−0.033, 0.016]	0.849	−0.016 [−0.049, 0.018]	0.820	0.018 [−0.011, 0.047]	0.902	−0.024 [−0.049, −0.002]
KO-F-position	0.913	−0.004 [−0.020, 0.011]	0.869	0.004 [−0.011, 0.020]	0.791	−0.011 [−0.036, 0.013]	0.918	−0.009 [−0.022, 0.002]
KO-F-neighbourhood	0.884	−0.033 [−0.056, −0.011]	0.771	−0.093 [−0.127, −0.062]	0.740	−0.062 [−0.096, −0.031]	0.871	−0.056 [−0.082, −0.031]
KO-F-rank	0.889	−0.029 [−0.056, −0.004]	0.851	−0.013 [−0.040, 0.013]	0.773	−0.029 [−0.064, 0.007]	0.882	−0.044 [−0.069, −0.020]
KO-F-item	0.916	−0.002 [−0.018, 0.011]	0.871	0.007 [−0.011, 0.024]	0.784	−0.018 [−0.047, 0.009]	0.927	0.000 [−0.011, 0.011]

Differences are variant minus reference; positive favours the variant. Each row is one refit seed with the configuration chosen by the design-item search; the intervals (95%, paired percentile bootstrap, 10 000 resamples) cover item sampling only. KO-ALL is a control (decisions identical to the unweighted rule on every item) and has no interval. Nothing is bold; rows in the order of the plan. Source: runs/P2-LRF-ABL-SCORE/ablation_analysis.json; script paper2-tnnls/scripts/tab_ablation.py.

TABLE S10
Exploratory Ablation: Accuracy and Difference From the Frozen LRF, LG Cells (the First 300 L1 Images)

	clean ($n = 300$)		shot ($n = 300$)		impulse ($n = 300$)		speckle ($n = 300$)	
Row	Acc.	d vs LRF [95 %]	Acc.	d vs LRF [95 %]	Acc.	d vs LRF [95 %]	Acc.	d vs LRF [95 %]
NONE (reference)	0.890	–	0.673	–	0.700	–	0.697	–
LRF (reference)	0.913	–	0.767	–	0.750	–	0.757	–
A-LIN	0.917	0.003 [−0.017, 0.023]	0.737	−0.030 [−0.057, −0.007]	0.753	0.003 [−0.023, 0.033]	0.753	−0.003 [−0.033, 0.027]
A-S-BD	0.920	0.007 [−0.017, 0.033]	0.773	0.007 [−0.030, 0.043]	0.793	0.043 [0.003, 0.083]	0.773	0.017 [−0.023, 0.057]
A-S-s3	0.893	−0.020 [−0.050, 0.010]	0.750	−0.017 [−0.060, 0.023]	0.743	−0.007 [−0.053, 0.037]	0.757	0.000 [−0.043, 0.043]
A-S-tbv	0.900	−0.013 [−0.043, 0.017]	0.730	−0.037 [−0.083, 0.007]	0.747	−0.003 [−0.050, 0.043]	0.753	−0.003 [−0.043, 0.033]
A-S-ltex	0.893	−0.020 [−0.053, 0.010]	0.673	−0.093 [−0.143, −0.047]	0.713	−0.037 [−0.087, 0.013]	0.683	−0.073 [−0.117, −0.030]
A-S-nbr	0.893	−0.020 [−0.050, 0.010]	0.710	−0.057 [−0.107, −0.010]	0.690	−0.060 [−0.110, −0.010]	0.720	−0.037 [−0.080, 0.007]
A-S-pos	0.900	−0.013 [−0.047, 0.020]	0.700	−0.067 [−0.110, −0.023]	0.737	−0.013 [−0.060, 0.033]	0.727	−0.030 [−0.070, 0.010]
A-SF	0.883	−0.030 [−0.063, 0.003]	0.700	−0.067 [−0.113, −0.020]	0.727	−0.023 [−0.073, 0.027]	0.730	−0.027 [−0.067, 0.010]
A-MIL	0.857	−0.057 [−0.093, −0.023]	0.760	−0.007 [−0.050, 0.037]	0.763	0.013 [−0.030, 0.057]	0.780	0.023 [−0.023, 0.070]
A-UNL	0.890	−0.023 [−0.050, 0.003]	0.673	−0.093 [−0.143, −0.047]	0.700	−0.050 [−0.100, 0.000]	0.697	−0.060 [−0.103, −0.020]
KO-S-BD	0.903	−0.010 [−0.033, 0.013]	0.703	−0.063 [−0.100, −0.027]	0.720	−0.030 [−0.067, 0.007]	0.720	−0.037 [−0.067, −0.007]
KO-S-s3	0.917	0.003 [−0.013, 0.020]	0.760	−0.007 [−0.033, 0.017]	0.760	0.010 [−0.013, 0.033]	0.777	0.020 [−0.003, 0.043]
KO-S-tbv	0.903	−0.010 [−0.027, 0.007]	0.730	−0.037 [−0.067, −0.007]	0.750	0.000 [−0.030, 0.030]	0.740	−0.017 [−0.047, 0.010]
KO-S-ltex	0.907	−0.007 [−0.027, 0.010]	0.737	−0.030 [−0.057, −0.007]	0.737	−0.013 [−0.033, 0.007]	0.767	0.010 [−0.010, 0.033]
KO-S-nbr	0.917	0.003 [−0.020, 0.027]	0.790	0.023 [−0.010, 0.060]	0.783	0.033 [−0.007, 0.073]	0.783	0.027 [−0.013, 0.067]
KO-S-pos	0.913	0.000 [−0.017, 0.017]	0.763	−0.003 [−0.030, 0.023]	0.750	0.000 [−0.027, 0.027]	0.750	−0.007 [−0.030, 0.017]
KO-F-token	0.900	−0.013 [−0.043, 0.017]	0.773	0.007 [−0.027, 0.040]	0.770	0.020 [−0.017, 0.057]	0.770	0.013 [−0.020, 0.047]
KO-F-position	0.913	0.000 [−0.017, 0.017]	0.763	−0.003 [−0.030, 0.023]	0.750	0.000 [−0.027, 0.027]	0.750	−0.007 [−0.030, 0.017]
KO-F-neighbourhood	0.863	−0.050 [−0.080, −0.020]	0.697	−0.070 [−0.110, −0.033]	0.717	−0.033 [−0.077, 0.007]	0.717	−0.040 [−0.077, −0.003]
KO-F-rank	0.877	−0.037 [−0.067, −0.007]	0.727	−0.040 [−0.080, 0.000]	0.753	0.003 [−0.043, 0.050]	0.723	−0.033 [−0.073, 0.007]
KO-F-item	0.917	0.003 [−0.013, 0.020]	0.733	−0.033 [−0.070, 0.000]	0.747	−0.003 [−0.040, 0.033]	0.737	−0.020 [−0.050, 0.010]

As Table S9. The LG cells use the first 300 images of the L1 cells, so the cells are not independent samples.

denoised unweighted sum and −0.051 (−0.089 to −0.013; 29 against 52) for the denoised LRF; against the undenoised unweighted sum they were 0.004 (−0.031 to 0.038) and −0.036 (−0.078 to 0.004), and the denoised LRF was below the denoised unweighted sum by 0.040 (−0.069 to −0.011; 13 against 31). Intervals are 95% paired percentile bootstrap intervals over items, unadjusted, from one resampling matrix per condition shared by all contrasts; there is no test, and the items were already used for the confirmatory test. The corrupted images were regenerated for this run; their tokenisation agreed with the stored L1 codes on 99.8% of image tokens under noise and 99.7% under JPEG (all 1 024 tokens equal on 69 and 73 of the 450 images). Source: runs/P2-LRF-DN; script paper2-tnnls/scripts/tab_baselines.py.

Deviations entry: Ablation (exploratory; added after the confirmatory results): plan research/paper2/tnnls/ABLATION-PLAN-2026-10-30.md committed before any variant was scored on fresh items; variants fitted only on the 500 design items; every variant reported; intervals only.

TABLE S11
EXPLORATORY ABLATION: DIFFERENCE FROM THE UNWEIGHTED RULE [95% INTERVAL], ALL CELLS

	L1 cells ($n = 450$)				LG cells ($n = 300$)			
Row	clean	JPEG 10	noise 0.1	splice 128	clean	shot	impulse	speckle
A-LIN	0.033 [0.009, 0.060]	0.036 [0.007, 0.064]	0.058 [0.020, 0.096]	0.036 [0.011, 0.062]	0.027 [0.000, 0.057]	0.063 [0.017, 0.110]	0.053 [0.000, 0.107]	0.057 [0.013, 0.100]
A-S-BD	0.036 [0.013, 0.058]	0.022 [0.000, 0.044]	0.067 [0.033, 0.100]	0.016 [−0.009, 0.040]	0.030 [0.007, 0.057]	0.100 [0.057, 0.147]	0.093 [0.053, 0.137]	0.077 [0.033, 0.120]
A-S-s3	0.007 [−0.011, 0.024]	0.004 [−0.016, 0.024]	0.036 [0.007, 0.064]	−0.007 [−0.024, 0.011]	0.003 [−0.017, 0.023]	0.077 [0.040, 0.113]	0.043 [0.013, 0.077]	0.060 [0.023, 0.097]
A-S-tbv	0.027 [0.004, 0.049]	0.007 [−0.020, 0.033]	0.029 [0.007, 0.053]	0.009 [−0.013, 0.031]	0.010 [−0.013, 0.037]	0.057 [0.027, 0.090]	0.047 [0.020, 0.077]	0.057 [0.020, 0.093]
A-S-ltex	0.009 [−0.016, 0.033]	−0.009 [−0.038, 0.018]	0.024 [−0.002, 0.053]	−0.002 [−0.024, 0.020]	0.003 [−0.027, 0.030]	0.000 [−0.010, 0.010]	0.013 [−0.010, 0.040]	−0.013 [−0.033, 0.003]
A-S-nbr	0.013 [−0.007, 0.033]	0.024 [0.000, 0.049]	−0.009 [−0.033, 0.016]	0.002 [−0.020, 0.024]	0.003 [−0.020, 0.030]	0.037 [0.007, 0.070]	−0.010 [−0.043, 0.023]	0.023 [−0.007, 0.057]
A-S-pos	0.016 [−0.013, 0.044]	0.016 [−0.013, 0.047]	−0.002 [−0.036, 0.031]	−0.004 [−0.036, 0.024]	0.010 [−0.023, 0.043]	0.027 [−0.010, 0.067]	0.037 [−0.003, 0.077]	0.030 [−0.007, 0.067]
A-SF	0.007 [−0.022, 0.036]	−0.009 [−0.040, 0.022]	0.009 [−0.024, 0.042]	−0.004 [−0.036, 0.027]	−0.007 [−0.040, 0.027]	0.027 [−0.013, 0.067]	0.027 [−0.017, 0.070]	0.033 [−0.007, 0.073]
A-MIL	−0.022 [−0.056, 0.011]	−0.011 [−0.049, 0.027]	0.060 [0.018, 0.104]	−0.044 [−0.080, −0.009]	−0.033 [−0.073, 0.007]	0.087 [0.030, 0.143]	0.063 [0.007, 0.120]	0.083 [0.030, 0.137]
A-UNL	0.009 [−0.011, 0.029]	−0.011 [−0.036, 0.011]	−0.002 [−0.024, 0.020]	−0.004 [−0.024, 0.016]	0.000 [−0.023, 0.023]	0.000 [−0.023, 0.023]	0.000 [−0.030, 0.030]	0.000 [−0.030, 0.030]
KO-S-BD	0.022 [−0.007, 0.051]	−0.013 [−0.044, 0.018]	0.016 [−0.018, 0.051]	0.011 [−0.018, 0.040]	0.013 [−0.020, 0.047]	0.030 [−0.010, 0.073]	0.020 [−0.023, 0.063]	0.023 [−0.017, 0.060]
KO-S-s3	0.027 [0.002, 0.053]	0.027 [−0.002, 0.056]	0.078 [0.040, 0.118]	0.022 [−0.002, 0.047]	0.027 [0.000, 0.057]	0.087 [0.040, 0.133]	0.060 [0.013, 0.107]	0.080 [0.037, 0.123]
KO-S-tbv	0.018 [−0.009, 0.044]	0.031 [0.002, 0.060]	0.051 [0.013, 0.089]	0.004 [−0.024, 0.033]	0.013 [−0.017, 0.047]	0.057 [0.013, 0.103]	0.050 [0.003, 0.097]	0.043 [0.000, 0.087]
KO-S-ltex	0.024 [−0.002, 0.051]	−0.027 [−0.060, 0.007]	0.053 [0.016, 0.091]	0.020 [−0.007, 0.049]	0.017 [−0.017, 0.050]	0.063 [0.017, 0.110]	0.037 [−0.007, 0.080]	0.070 [0.023, 0.117]
KO-S-nbr	0.040 [0.013, 0.067]	0.013 [−0.016, 0.042]	0.109 [0.071, 0.149]	0.029 [0.002, 0.056]	0.027 [−0.003, 0.060]	0.117 [0.067, 0.167]	0.083 [0.033, 0.133]	0.087 [0.037, 0.137]
KO-S-pos	0.027 [0.002, 0.051]	0.020 [−0.009, 0.047]	0.058 [0.022, 0.096]	0.022 [−0.002, 0.047]	0.023 [−0.003, 0.053]	0.090 [0.043, 0.140]	0.050 [0.007, 0.093]	0.053 [0.013, 0.093]
KO-F-token	0.022 [−0.007, 0.051]	0.000 [−0.029, 0.029]	0.087 [0.051, 0.122]	0.007 [−0.022, 0.036]	0.010 [−0.023, 0.043]	0.100 [0.057, 0.143]	0.070 [0.027, 0.113]	0.073 [0.030, 0.117]
KO-F-position	0.027 [0.002, 0.051]	0.020 [−0.009, 0.047]	0.058 [0.022, 0.096]	0.022 [−0.002, 0.047]	0.023 [−0.003, 0.053]	0.090 [0.043, 0.140]	0.050 [0.007, 0.093]	0.053 [0.013, 0.093]
KO-F-neighbourhood	−0.002 [−0.036, 0.031]	−0.078 [−0.118, −0.040]	0.007 [−0.031, 0.044]	−0.024 [−0.058, 0.007]	−0.027 [−0.067, 0.013]	0.023 [−0.023, 0.070]	0.017 [−0.033, 0.067]	0.020 [−0.027, 0.067]
KO-F-rank	0.002 [−0.027, 0.029]	0.002 [−0.029, 0.033]	0.040 [0.002, 0.078]	−0.013 [−0.042, 0.016]	−0.013 [−0.047, 0.020]	0.053 [0.010, 0.100]	0.053 [0.007, 0.100]	0.027 [−0.020, 0.073]
KO-F-item	0.029 [0.004, 0.053]	0.022 [−0.007, 0.051]	0.051 [0.020, 0.082]	0.031 [0.007, 0.056]	0.027 [0.000, 0.057]	0.060 [0.023, 0.100]	0.047 [0.013, 0.080]	0.040 [0.003, 0.077]

Variant minus the unweighted rule; positive favours the variant. The interval of LRF against the unweighted rule is the preregistered one (main paper, table of preregistered families) and is not recomputed here.

TABLE S12
EXPLORATORY ABLATION: SEED-SPREAD REFITS (ACCURACY AT SEEDS 0, 1, 2; MEAN, SD, RANGE)

	L1 cells ($n = 450$)				LG cells ($n = 300$)			
Refit	clean	JPEG 10	noise 0.1	splice 128	clean	shot	impulse	speckle
LRF refit (harness)	0.918, 0.900, 0.913 0.910, 0.009, 0.018	0.864, 0.844, 0.838 0.849, 0.014, 0.027	0.802, 0.787, 0.780 0.790, 0.011, 0.022	0.927, 0.904, 0.916 0.916, 0.011, 0.022	0.913, 0.893, 0.910 0.906, 0.011, 0.020	0.767, 0.747, 0.737 0.750, 0.015, 0.030	0.750, 0.757, 0.760 0.756, 0.005, 0.010	0.757, 0.760, 0.760 0.759, 0.002, 0.003
A-LIN	0.920, 0.920, 0.920 0.920, 0.000, 0.000	0.884, 0.869, 0.869 0.874, 0.009, 0.016	0.791, 0.787, 0.782 0.787, 0.004, 0.009	0.931, 0.927, 0.924 0.927, 0.003, 0.007	0.917, 0.913, 0.910 0.913, 0.003, 0.007	0.737, 0.743, 0.747 0.742, 0.005, 0.010	0.753, 0.757, 0.747 0.752, 0.005, 0.010	0.753, 0.753, 0.757 0.754, 0.002, 0.003
A-S-BD	0.922, 0.920, 0.920 0.921, 0.001, 0.002	0.871, 0.873, 0.871 0.872, 0.001, 0.002	0.800, 0.802, 0.802 0.801, 0.001, 0.002	0.911, 0.916, 0.913 0.913, 0.002, 0.004	0.920, 0.913, 0.913 0.916, 0.004, 0.007	0.773, 0.793, 0.787 0.784, 0.010, 0.020	0.793, 0.797, 0.793 0.794, 0.002, 0.003	0.773, 0.783, 0.777 0.778, 0.005, 0.010
A-S-s3	0.893, 0.896, 0.896 0.895, 0.001, 0.002	0.853, 0.847, 0.851 0.850, 0.003, 0.007	0.769, 0.776, 0.778 0.774, 0.005, 0.009	0.889, 0.889, 0.887 0.888, 0.001, 0.002	0.893, 0.900, 0.897 0.897, 0.003, 0.007	0.750, 0.747, 0.750 0.749, 0.002, 0.003	0.743, 0.743, 0.747 0.744, 0.002, 0.003	0.757, 0.757, 0.757 0.757, 0.000, 0.000
A-S-tbv	0.913, 0.907, 0.909 0.910, 0.003, 0.007	0.856, 0.844, 0.844 0.848, 0.006, 0.011	0.762, 0.767, 0.762 0.764, 0.003, 0.004	0.904, 0.913, 0.907 0.908, 0.005, 0.009	0.900, 0.910, 0.907 0.906, 0.005, 0.010	0.730, 0.730, 0.720 0.727, 0.006, 0.010	0.747, 0.753, 0.753 0.751, 0.004, 0.007	0.753, 0.773, 0.763 0.763, 0.010, 0.020
A-S-ltex	0.896, 0.893, 0.893 0.894, 0.001, 0.002	0.840, 0.838, 0.838 0.839, 0.001, 0.002	0.758, 0.751, 0.753 0.754, 0.003, 0.007	0.893, 0.891, 0.896 0.893, 0.002, 0.004	0.893, 0.890, 0.887 0.890, 0.003, 0.007	0.673, 0.673, 0.667 0.671, 0.004, 0.007	0.713, 0.723, 0.723 0.720, 0.006, 0.010	0.683, 0.683, 0.683 0.683, 0.000, 0.000
A-S-nbr	0.900, 0.902, 0.902 0.901, 0.001, 0.002	0.873, 0.876, 0.876 0.875, 0.001, 0.002	0.724, 0.724, 0.722 0.724, 0.001, 0.002	0.898, 0.900, 0.900 0.899, 0.001, 0.002	0.893, 0.900, 0.897 0.897, 0.003, 0.007	0.710, 0.710, 0.710 0.710, 0.000, 0.000	0.690, 0.693, 0.693 0.692, 0.002, 0.003	0.720, 0.723, 0.723 0.722, 0.002, 0.003
A-S-pos	0.902, 0.907, 0.896 0.901, 0.006, 0.011	0.864, 0.869, 0.858 0.864, 0.006, 0.011	0.731, 0.731, 0.742 0.735, 0.006, 0.011	0.891, 0.893, 0.889 0.891, 0.002, 0.004	0.900, 0.900, 0.887 0.896, 0.008, 0.013	0.700, 0.700, 0.707 0.702, 0.004, 0.007	0.737, 0.740, 0.737 0.738, 0.002, 0.003	0.727, 0.727, 0.727 0.727, 0.000, 0.000
A-SF	0.893, 0.893, 0.896 0.894, 0.001, 0.002	0.840, 0.840, 0.836 0.839, 0.003, 0.004	0.742, 0.740, 0.744 0.742, 0.002, 0.004	0.891, 0.891, 0.889 0.890, 0.001, 0.002	0.883, 0.883, 0.887 0.884, 0.002, 0.003	0.700, 0.700, 0.700 0.700, 0.000, 0.000	0.727, 0.727, 0.727 0.727, 0.000, 0.000	0.730, 0.730, 0.730 0.730, 0.000, 0.000
A-MIL	0.864, 0.860, 0.864 0.863, 0.003, 0.004	0.838, 0.833, 0.838 0.836, 0.003, 0.004	0.793, 0.807, 0.809 0.803, 0.008, 0.016	0.851, 0.853, 0.869 0.858, 0.010, 0.018	0.857, 0.863, 0.857 0.859, 0.004, 0.007	0.760, 0.760, 0.770 0.763, 0.006, 0.010	0.763, 0.773, 0.777 0.771, 0.007, 0.013	0.780, 0.767, 0.777 0.774, 0.007, 0.013

Seed 0 is the pre-fixed seed, not the best of three; the extra seeds repeat only the final fit with the configuration of the seed-0 search, so they understate refit variation. "LRF refit (harness)" is a seed-spread refit, not the frozen LRF. Descriptive; no interval.

TABLE S13
EXPLORATORY COMPARISON OF WAYS OF COMBINING THE SAME PER-TOKEN EVIDENCE: ACCURACY AND DIFFERENCE FROM THE UNWEIGHTED SUM ON THE 450 FRESH-FOLD IMAGES

	Noise 0.1		JPEG 10		Splice 128		Clean	
Method	Acc.	Diff. from unweighted (95% interval)	Acc.	Diff. from unweighted (95% interval)	Acc.	Diff. from unweighted (95% interval)	Acc.	Diff. from unweighted (95% interval)
Unweighted sum†	0.733	–	0.849	–	0.896	–	0.887	–
Unlearned rank weight	0.731	−0.002 [−0.024, 0.020]	0.838	−0.011 [−0.036, 0.011]	0.891	−0.004 [−0.024, 0.016]	0.896	+0.009 [−0.011, 0.029]
Median	0.716	−0.018 [−0.038, 0.002]	0.829	−0.020 [−0.040, 0.000]	0.864	−0.031 [−0.051, −0.013]	0.862	−0.024 [−0.042, −0.007]
20% trimmed mean	0.716	−0.018 [−0.033, −0.004]	0.827	−0.022 [−0.040, −0.007]	0.873	−0.022 [−0.040, −0.004]	0.869	−0.018 [−0.033, −0.004]
Dempster combination	0.722	−0.011 [−0.036, 0.013]	0.838	−0.011 [−0.033, 0.011]	0.889	−0.007 [−0.027, 0.016]	0.884	−0.002 [−0.024, 0.020]
Single feature (selected)	0.742	+0.009 [−0.024, 0.042]	0.840	−0.009 [−0.040, 0.022]	0.891	−0.004 [−0.036, 0.027]	0.893	+0.007 [−0.022, 0.036]
Linear logit, 35 features	0.791	+0.058 [0.020, 0.096]	*0.884*	+0.036 [0.007, 0.064]	*0.931*	+0.036 [0.011, 0.062]	0.920	+0.033 [0.009, 0.060]
Additive, B and D only	0.800	+0.067 [0.033, 0.100]	0.871	+0.022 [0.000, 0.044]	0.911	+0.016 [−0.009, 0.040]	*0.922*	+0.036 [0.013, 0.058]
Attention pooling (MIL)	0.793	+0.060 [0.018, 0.104]	0.838	−0.011 [−0.049, 0.027]	0.851	−0.044 [−0.080, −0.009]	0.864	−0.022 [−0.056, 0.011]
LRF (frozen)†	0.802	+0.069 [0.029, 0.109]	0.864	+0.016 [−0.013, 0.044]	0.927	+0.031 [0.007, 0.058]	0.918	+0.031 [0.004, 0.058]
Denoise, unweighted sum	0.844	+0.111 [0.067, 0.156]	0.853	+0.004 [−0.031, 0.038]	–	–	–	–
Denoise, LRF	*0.856*	+0.122 [0.071, 0.173]	0.813	−0.036 [−0.078, 0.004]	–	–	–	–

As Table IV of the main paper, with the difference from the unweighted sum (accuracy minus the unweighted sum; 95% paired percentile bootstrap interval, 10 000 resamples). Intervals are marginal and unadjusted; the rows other than the unweighted sum and LRF are exploratory, and the intervals of the ablation rows, the median and trimmed mean, and the denoise rows come from separate runs with their own resampling seeds. †Fixed before the fold was opened. Source: runs/P2-LRF-L1/a6_analysis.json, runs/P2-LRF-ABL-SCORE/ablation_analysis.json, runs/P2-LRF-E2/combiners.json, runs/P2-LRF-E4/evidential.json; script paper2-tnnls/scripts/tab_baselines.py.

TABLE S14
EXPLORATORY GAUSSIAN-NOISE SEVERITY SWEEP OF THE FROZEN LRF AND THE UNWEIGHTED SUM ON THE 450 FRESH-FOLD IMAGES

σ	Unweighted	LRF	LRF − unw.	95% interval	b/c	Gain − clean gain [95%]
0 (clean)	0.887	0.918	+0.031	[0.004, 0.058]	27/13	–
0.05	0.793	0.849	+0.056	[0.020, 0.091]	46/21	+0.024 [−0.016, 0.064]
0.1 (fitted level)	0.733	0.802	+0.069	[0.029, 0.109]	58/27	+0.038 [−0.004, 0.080]
0.2	0.649	0.707	+0.058	[0.018, 0.098]	57/31	+0.027 [−0.020, 0.076]
0.3	0.600	0.684	+0.084	[0.044, 0.124]	62/24	+0.053 [0.009, 0.098]

Exploratory, planned before the run (research/paper2/tnnls/EXPLORATORY-PLAN-SEVERITY-2026-10-31.md); no test. σ on $[0, 1]$ pixel values; each image has one noise draw (the stream of the confirmatory noise cell), scaled by σ, so the levels differ in severity only. Levels 0 and 0.1 are the confirmatory run's cells; the others come from runs/P2-LRF-SEV. Rebuilding the $\sigma = 0.1$ images with the same draw reproduced the stored image tokens at 99.8% agreement (threshold 99%). b/c: images that LRF decided correctly and the unweighted sum wrongly, and the reverse. Gain − clean gain: the per-image difference of (LRF − unweighted) between the noisy and the clean image. 95% paired percentile bootstrap intervals, 10 000 resamples, one resampling of images shared by all levels; marginal and unadjusted. LRF was fitted at $\sigma = 0.1$ only. For comparison with a corruption benchmark for image-text models, its Gaussian levels are 0.08 to 0.38 (Qiu et al., cited in the main paper).

S-VI. SOURCES OF THE MAIN PAPER'S FIGURES AND TABLES

Every figure and table of the main paper is computed by a script in paper2-tnnls/scripts/ that reads only committed run files.

- Fig. 1 (conceptual, no data): script paper2-tnnls/scripts/fig1_measurement.py; definitions in Section III.
- Feature groups (Fig. 2 of the main paper): runs/P2-LRF-FROZEN-LFIX/spec.json (features); analysis_posthoc/learned_ceiling/data.py, functions feature_names and engineer; full table in Table S7.
- Fig. 2 (frozen weight model, schematic) and Fig. 5 (partial dependence): runs/P2-LRF-FROZEN-LFIX/{model.npz, spec.json}; computed values in runs/P2-PAPER-FIGS/ (fig4_meta.json, fig4_pd_curves.json); script paper2-tnnls/scripts/fig4_study2_weights.py. Each curve of Fig. 5 runs between the lowest and highest of the eight knots of its feature; the effect range is the range of w over the plotted domain.
- Fig. 6 (exploratory severity sweep): runs/P2-LRF-SEV/sev_analysis.json, keys levels.*.{NONE,LRF,LRF_minus_NONE}; plotted values in runs/P2-PAPER-FIGS/fig_severity.json; script paper2-tnnls/scripts/fig_severity.py.
- Fig. 4 (three fresh-fold items): runs/P2-PAPER-FIGS/fig_qualitative.json (selection rule, items, colour scales, attribution); runs/P2-LRF-L1/predictions.jsonl (y_none, y_lrf, condition noise0.1); runs/P2-LRF-L1/arrays/item000{0,1,8}.npz (SHA-256 in the figure's trace file); the noisy pixels are regenerated with the runner's corruption call and seeds; colour scales are shared by the rows, Δ_j is clipped at the 99th percentile of $|\Delta_j|$, and the weights are on a square-root scale; script paper2-tnnls/scripts/fig_qualitative.py. Images: COCO val2017 (Lin et al., 2014; cited in the main paper), Flickr, licences as listed; the noisy panels are modified. Image 335177: Attribution-NonCommercial-ShareAlike (CC BY-NC-SA 2.0), http://farm4.staticflickr.com/3444/3821042011_c741061caa_z.jpg; image 520324: Attribution-ShareAlike (CC BY-SA 2.0), http://farm2.staticflickr.com/1254/5130147192_a2f29e8ea1_z.jpg; image 224724: Attribution-NonCommercial (CC BY-NC 2.0), http://farm4.staticflickr.com/3543/3544616491_7cdec47006_z.jpg (record: runs/P2-PAPER-FIGS/fig_qualitative_licences.json).
- Algorithm 1 (decision rule of the frozen LRF): runs/P2-LRF-FROZEN-LFIX/spec.json (family, n_features, caption_slot_len, preprocessing).

- Table IV (comparison with alternative aggregation rules, exploratory) and Table S13: runs/P2-LRF-L1/a6_analysis.json, runs/P2-LRF-ABL-SCORE/ablation_analysis.json, runs/P2-LRF-E2/combiners.json, runs/P2-LRF-E4/evidential.json (runs/P2-LRF-DN/dn_analysis.json for the denoise rows); script paper2-tnnls/scripts/tab_baselines.py.
- Table I (preregistered families): runs/P2-LRF-L1/a6_analysis.json (family_L.*); runs/P2-LRF-LG/a6_analysis.json (family_LG.cells.*).
- Table II (Flickr30K transfer test, F1, F-NI, F-LEN): runs/P2-LRF-F30K/f30k_analysis.json (F1, F_NI, F_LEN); script paper2-tnnls/scripts/tab_f30k.py.
- Fig. 3 (family L and L-NI): runs/P2-LRF-L1/a6_analysis.json, keys family_L.*; script paper2-tnnls/scripts/fig5_study2_results.py.
- Figure S1 (item-level margins, descriptive): runs/P2-LRF-L1/predictions.jsonl (fields condition, idx, m_none, m_lrf, y_none, y_lrf; SHA-256 checked against a6_analysis.json); plotted values in runs/P2-PAPER-FIGS/fig_margins.json; script paper2-tnnls/scripts/fig_margins.py.
- Fig. S2 (exploratory ablation): runs/P2-LRF-ABL-SCORE/ablation_analysis.json, keys rows[*].cells.L1_noise0.1.vs_LRF.*, ko_all_control; plotted values in runs/P2-PAPER-FIGS/fig_ablation.json; script paper2-tnnls/scripts/fig_ablation.py; rows in the order of Section S-V.
- Table III (reference encoders): runs/P2-LRF-BASELINES/baselines.json (table.*); runs/P2-PAPER-TABLES/baselines_clean300.json; revisions in Section S-IV.
  Pixel parity: runs/P2-LRF-BASELINES-PARITY-L1 and runs/P2-LRF-BASELINES-PARITY-LG.